2.2.2 Derivation of the Verlet Method

Now let us consider the discrete version of "minimizing the action". We work on the time interval $[0, \tau]$. Consider the $\nu + 1$ points $\boldsymbol{q}_0$ to $\boldsymbol{q}_\nu$ in configuration space

$$\hat{\Gamma} = (\boldsymbol{q}_0, \boldsymbol{q}_1, \ldots, \boldsymbol{q}_\nu).$$

We must first define the action for such a discrete path. One perspective is that we first formulate a discrete version of the Lagrangian in terms of only the point set $\boldsymbol{q}_0, \boldsymbol{q}_1, \ldots \boldsymbol{q}_\nu$, but this somehow requires that we define velocities. Noting that

$$\dot{\boldsymbol{q}}(t) \approx \frac{\boldsymbol{q}(t+h) - \boldsymbol{q}(t)}{h},$$

we are led to consider the approximation at time level n

$$\boldsymbol{v}_n \stackrel{\text{def}}{=} \frac{\boldsymbol{q}_{n+1} - \boldsymbol{q}_n}{h},$$

and thus, by Riemann summation

$$\mathcal{A}_L \approx \hat{\mathcal{A}} \stackrel{\text{def}}{=} \sum_{n=0}^{\nu-1} L\left(\boldsymbol{q}_n, \frac{\boldsymbol{q}_{n+1} - \boldsymbol{q}_n}{h}\right) h.$$

At first glance, it looks like this is a crude approximation, since we have employed a one-sided difference, however, let us proceed anyway to see the implications of this choice. In the case of a mechanical system with Lagrangian $L(\boldsymbol{q}, \boldsymbol{v}) = \dot{\boldsymbol{v}}^T \boldsymbol{M} \boldsymbol{v}/2 + U(\boldsymbol{q})$, we then have

$$\hat{\mathcal{A}} = \sum_{n=0}^{\nu-1} \left[\frac{(\boldsymbol{q}_{n+1} - \boldsymbol{q}_n)^T \boldsymbol{M} (\boldsymbol{q}_{n+1} - \boldsymbol{q}_n)}{2h^2} - U(\boldsymbol{q}_n) \right] h.$$

We should think of $\hat{\mathcal{A}}$ as a function of all the positions $\boldsymbol{q}_0, \boldsymbol{q}_1, \ldots, \boldsymbol{q}_\nu$ defining the discrete path. Critical points of this function satisfy

$$\nabla \hat{\mathcal{A}} = 0,$$

where the gradient must be taken with respect to all configurational points on the path (and all coordinates). This condition leads to the equations

$$\frac{\partial \hat{\mathcal{A}}}{\partial \boldsymbol{q}_n} = 0, \qquad n = 1, \ldots, \nu,$$

since we think of the starting point $\boldsymbol{q}_0$ as fixed. To calculate the derivative of $\hat{\mathcal{A}}$ with respect to $\boldsymbol{q}_n$, $n = 1, 2, \ldots, \nu - 1$, note that only a few terms of the discrete action involve this configurational point, thus

$$\begin{aligned}\frac{\partial \hat{\mathcal{A}}}{\partial \boldsymbol{q}_n} &= \frac{\partial}{\partial \boldsymbol{q}_n}\left[\frac{1}{2h}\left((\boldsymbol{q}_n - \boldsymbol{q}_{n-1})^T \boldsymbol{M}(\boldsymbol{q}_n - \boldsymbol{q}_{n-1}) + (\boldsymbol{q}_{n+1} - \boldsymbol{q}_n)^T\right.\right.\\ &\qquad \left.\left. \times\ \boldsymbol{M}(\boldsymbol{q}_{n+1} - \boldsymbol{q}_n)\right) - U(\boldsymbol{q}_n)h\right]\\ &= \frac{1}{h}\left[\boldsymbol{M}(\boldsymbol{q}_n - \boldsymbol{q}_{n-1}) - M(\boldsymbol{q}_{n+1} - \boldsymbol{q}_n)\right] - \nabla U(\boldsymbol{q}_n)h,\end{aligned}$$

which yields the equations

$$\boldsymbol{M}(\boldsymbol{q}_{n+1} - 2\boldsymbol{q}_n + \boldsymbol{q}_{n-1}) = -h^2 \nabla U(\boldsymbol{q}_n), \qquad n = 1, 2, \ldots, \nu - 1. \tag{2.4}$$

We can think of

$$\frac{1}{h^2}(\boldsymbol{q}_{n+1} - 2\boldsymbol{q}_n + \boldsymbol{q}_{n-1})$$

as a centered finite difference approximation of the acceleration (the second derivative of position), thus the Eq. (2.4) is a direct discretization of Newton's equations.

Somewhat surprisingly, the one-sided approximation of velocities in the discrete Lagrangian has led to a symmetric discretization of the equations of motion. There is a slight issue of what to do at the endpoints. In the variational formulation we think of the endpoints of the curve as fixed points which means that we are effectively solving a boundary value problem and Eq. (2.4) tells us how to compute all the interior points along the path. In the case of an initial value problem (the usual issue in molecular dynamics) we do not know in advance the right endpoint value, but we assume that we can, in some way, calculate $\boldsymbol{q}_1$ (by a *starting procedure*) and so Eq. (2.4) defines $\boldsymbol{q}_2, \boldsymbol{q}_3, \ldots, \boldsymbol{q}_\nu$.

The method (2.4) is commonly referred to as *Störmer's rule*. It was used by the mathematician Störmer for calculations in the first decade of the 1900s. In molecular dynamics this method is referred to as the Verlet method since it was used by Verlet in his important 1967 paper [387].

The scheme is usually given in an alternative "velocity Verlet" form that takes a step from a given vector $\boldsymbol{q}_n$, $\boldsymbol{v}_n$ to $\boldsymbol{q}_{n+1}$, $\boldsymbol{v}_{n+1}$ by the sequence of operations

$$\boldsymbol{v}_{n+1/2} = \boldsymbol{v}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{F}_n, \tag{2.5}$$

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{v}_{n+1/2}, \tag{2.6}$$

$$\boldsymbol{v}_{n+1} = \boldsymbol{v}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\boldsymbol{F}_{n+1}, \tag{2.7}$$